

JW Ch 6 Lab, Part 1

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Instructions and time-saving hints. The purpose of this lab is to learn linear regression in R. You will reproduce the commands blocks in JW Section 6.5 in this Rmarkdown file. **Please do not retype all the R commands as shown in Section 6.5.** Instead, go to link to text, find Chapter 6 Lab text, and cut and paste the text into your Rmarkdown file. Then break up the R commands into the R chunks shown in the text. **Do not put all commands into the same block!**

After breaking the text into the correct chunks, you may knit the document. You will notice some problems when knitting. This is due to the fact that the pasted commands are meant for the command line, not Rmarkdown. You must find Rmarkdown conversions for these problems. One problem is the appearance of the `fix()` commands in the blocks, which displays dataframes in a different window. As a substitute for `fix()`, please use the `kable` command from library `knitr`, as in `knitr::kable()`. The results, however, will be too long to hand in! Use this only for your personal use. For your homework/lab submission, please instead use `head()` instead of `fix()`. The TA will be less burdened that way.

Since R chunks are independent pieces of code, you will sometimes produce errors when imitating the format in the book. For example, the `abline()` commands must be grouped in the same chunk as the initial `plot()` command.

Congratulations! When the document is successfully knitted, you should read Section 6.5 along with the Rmarkdown output. You should see the commands along with the output plots. Together, JW 6.5 and your Rmarkdown file will help you to learn linear regression in R. Note that the plotting is done with core R commands and not with the `ggplot2` package. Submit your Rmd file along with an unzipped PDF of the result before the deadline.

Chapter 6 Lab 1: Subset Selection Methods

Best Subset Selection

```
library(ISLR)
```

```
## Warning: package 'ISLR' was built under R version 4.3.3
```

```
head(Hitters)
```

	AtBat	Hits	HmRun	Runs	RBI	Walks	Years	CAtBat	CHits	CHmRun
## -Andy Allanson	293	66	1	30	29	14	1	293	66	1
## -Alan Ashby	315	81	7	24	38	39	14	3449	835	69
## -Alvin Davis	479	130	18	66	72	76	3	1624	457	63
## -Andre Dawson	496	141	20	65	78	37	11	5628	1575	225
## -Andres Galarrraga	321	87	10	39	42	30	2	396	101	12
## -Alfredo Griffin	594	169	4	74	51	35	11	4408	1133	19

```
##           CRuns CRBI CWalks League Division PutOuts Assists Errors
## -Andy Allanson      30   29    14      A        E    446     33    20
## -Alan Ashby        321  414   375      N        W    632     43    10
## -Alvin Davis        224  266   263      A        W    880     82    14
## -Andre Dawson       828  838   354      N        E    200     11     3
## -Andres Galarrraga   48   46    33      N        E   805     40     4
## -Alfredo Griffin    501  336   194      A        W   282    421    25
##           Salary NewLeague
## -Andy Allanson      NA        A
## -Alan Ashby        475.0      N
## -Alvin Davis        480.0      A
## -Andre Dawson       500.0      N
## -Andres Galarrraga   91.5      N
## -Alfredo Griffin    750.0      A
```

```
names(Hitters)
```

```
## [1] "AtBat"      "Hits"       "HmRun"      "Runs"       "RBI"        "Walks"
## [7] "Years"      "CAtBat"     "CHits"      "CHmRun"     "CRuns"      "CRBI"
## [13] "CWalks"     "League"     "Division"   "PutOuts"    "Assists"    "Errors"
## [19] "Salary"     "NewLeague"
```

```
dim(Hitters)
```

```
## [1] 322  20
```

```
sum(is.na(Hitters$Salary))
```

```
## [1] 59
```

```
Hitters=na.omit(Hitters)
dim(Hitters)
```

```
## [1] 263  20
```

```
sum(is.na(Hitters))
```

```
## [1] 0
```

```
library(leaps)
```

```
## Warning: package 'leaps' was built under R version 4.3.2
```

```
regfit.full=regsubsets(Salary~.,Hitters)
summary(regfit.full)
```

```
## Subset selection object
## Call: regsubsets.formula(Salary ~ ., Hitters)
## 19 Variables (and intercept)
```

```

##           Forced in Forced out
## AtBat      FALSE      FALSE
## Hits       FALSE      FALSE
## HmRun      FALSE      FALSE
## Runs       FALSE      FALSE
## RBI        FALSE      FALSE
## Walks      FALSE      FALSE
## Years      FALSE      FALSE
## CatBat     FALSE      FALSE
## CHits      FALSE      FALSE
## CHmRun     FALSE      FALSE
## CRuns      FALSE      FALSE
## CRBI       FALSE      FALSE
## CWalks     FALSE      FALSE
## LeagueN    FALSE      FALSE
## DivisionW  FALSE      FALSE
## PutOuts    FALSE      FALSE
## Assists    FALSE      FALSE
## Errors     FALSE      FALSE
## NewLeagueN FALSE      FALSE
## 1 subsets of each size up to 8
## Selection Algorithm: exhaustive
##           AtBat Hits HmRun Runs RBI Walks Years CatBat CHits CHmRun CRuns CRBI
## 1 ( 1 ) " " " " " " " " " " " " " " " " " " " " " "
## 2 ( 1 ) " " "*" " " " " " " " " " " " " " " " "
## 3 ( 1 ) " " "*" " " " " " " " " " " " " " " " "
## 4 ( 1 ) " " "*" " " " " " " " " " " " " " " " "
## 5 ( 1 ) "*" "*" " " " " " " " " " " " " " " " "
## 6 ( 1 ) "*" "*" " " " " " " "*" " " " " " " " "
## 7 ( 1 ) " " "*" " " " " " " "*" " " "*" "*" " " "
## 8 ( 1 ) "*" "*" " " " " " " "*" " " " " "*" "*" " "
##           CWalks LeagueN DivisionW PutOuts Assists Errors NewLeagueN
## 1 ( 1 ) " " " " " " " " " " " "
## 2 ( 1 ) " " " " " " " " " " " "
## 3 ( 1 ) " " " " " " "*" " " " " "
## 4 ( 1 ) " " " " "*" "*" " " " " " "
## 5 ( 1 ) " " " " "*" "*" " " " " " "
## 6 ( 1 ) " " " " "*" "*" " " " " " "
## 7 ( 1 ) " " " " "*" "*" " " " " " "
## 8 ( 1 ) "*" " " "*" "*" " " " " " "

```

```

regfit.full=regsubsets(Salary~.,data=Hitters,nvmax=19)
reg.summary=summary(regfit.full)

```

```

names(reg.summary)

```

```

## [1] "which" "rsq" "rss" "adjr2" "cp" "bic" "outmat" "obj"

```

```

reg.summary$rsq

```

```

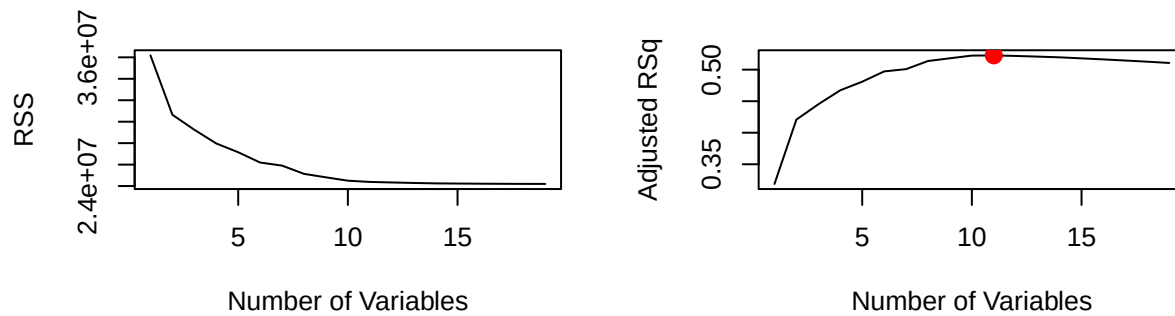
## [1] 0.3214501 0.4252237 0.4514294 0.4754067 0.4908036 0.5087146 0.5141227
## [8] 0.5285569 0.5346124 0.5404950 0.5426153 0.5436302 0.5444570 0.5452164
## [15] 0.5454692 0.5457656 0.5459518 0.5460945 0.5461159

```

```
par(mfrow=c(2,2))
plot(reg.summary$rss,xlab="Number of Variables",ylab="RSS",type="l")
plot(reg.summary$adjr2,xlab="Number of Variables",ylab="Adjusted RSq",type="l")
which.max(reg.summary$adjr2)
```

```
## [1] 11
```

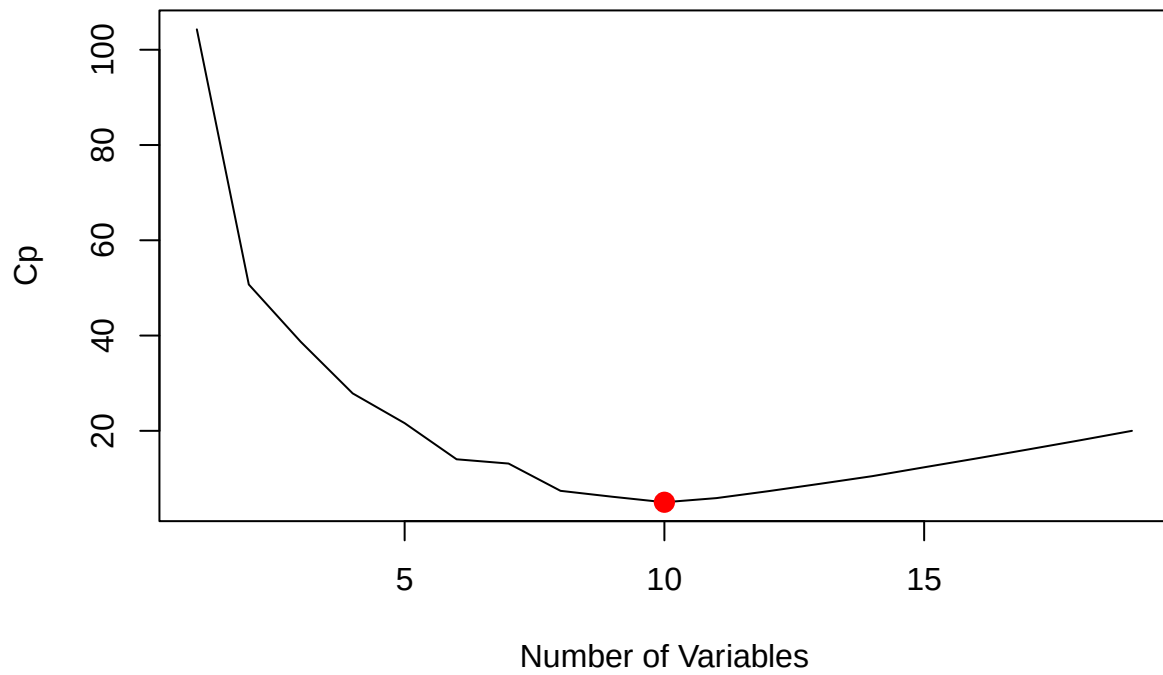
```
points(11,reg.summary$adjr2[11], col="red",cex=2,pch=20)
```



```
plot(reg.summary$cp,xlab="Number of Variables",ylab="Cp",type='l')
which.min(reg.summary$cp)
```

```
## [1] 10
```

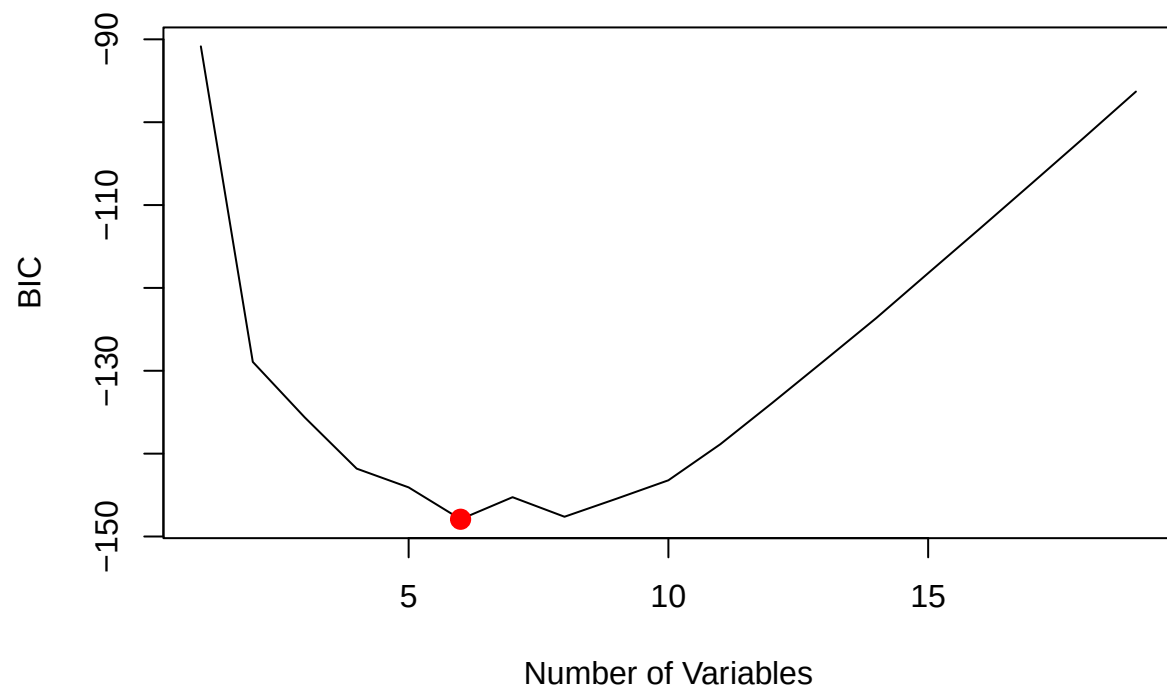
```
points(10,reg.summary$cp[10], col="red",cex=2,pch=20)
```



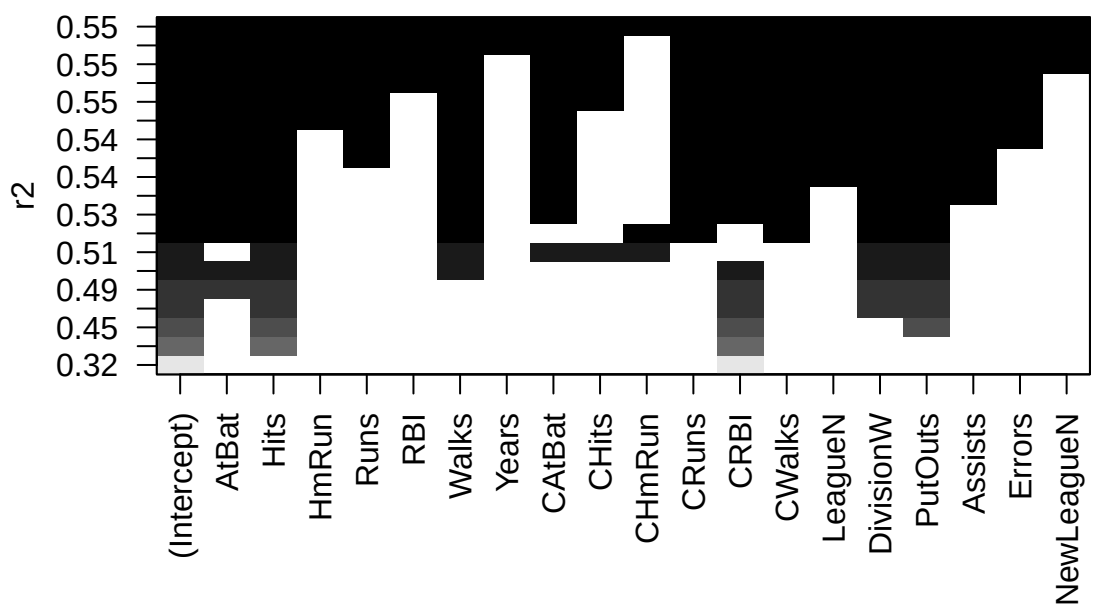
```
which.min(reg.summary$bic)
```

```
## [1] 6
```

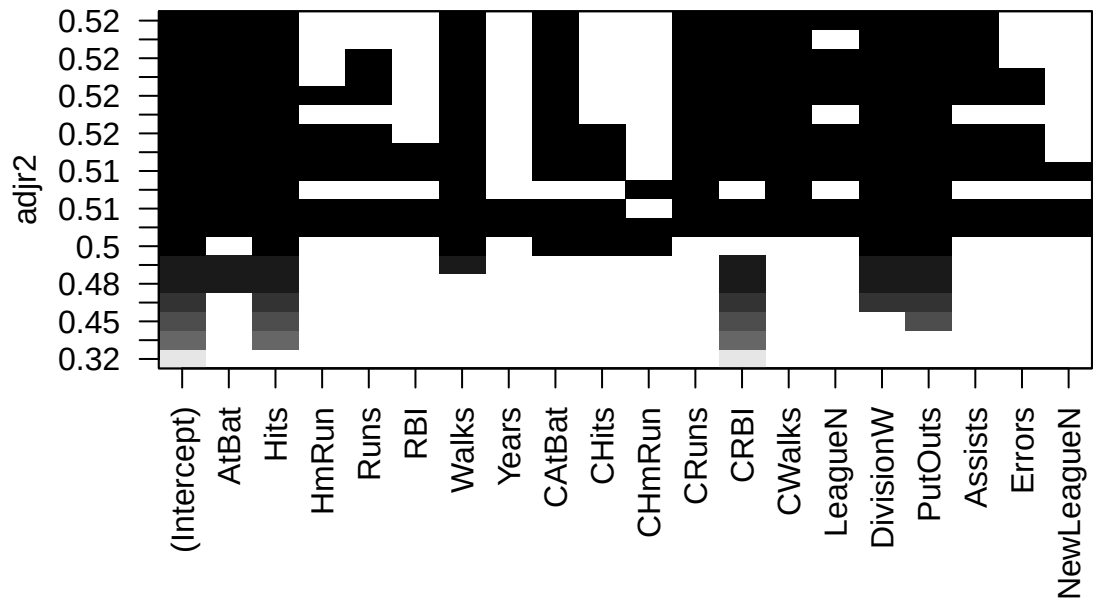
```
plot(reg.summary$bic,xlab="Number of Variables",ylab="BIC",type='l')  
points(6,reg.summary$bic[6],col="red",cex=2,pch=20)
```



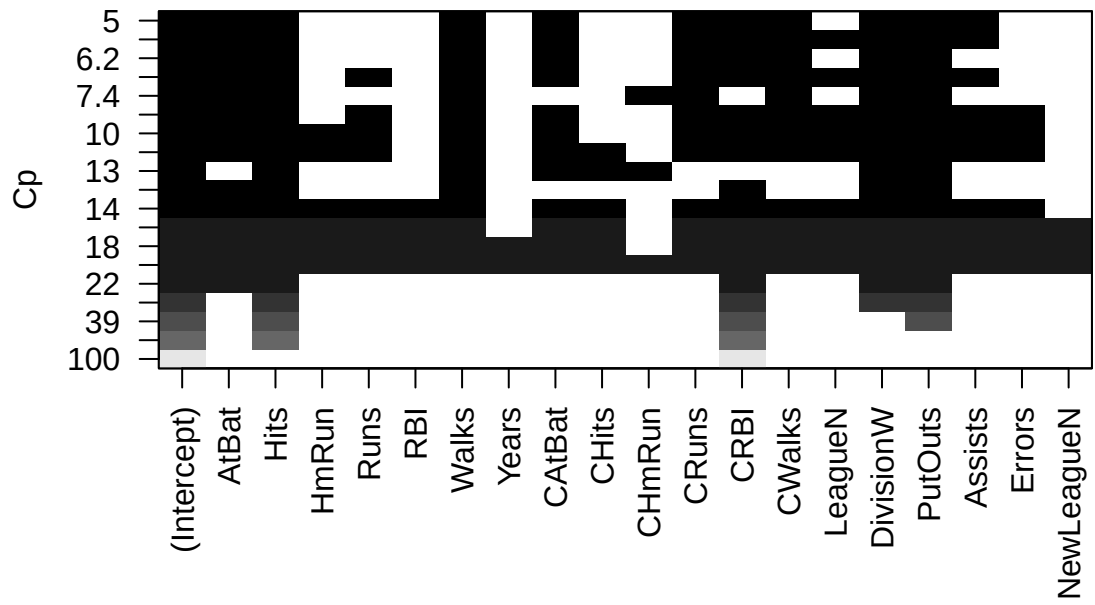
```
plot(regfit.full,scale="r2")
```



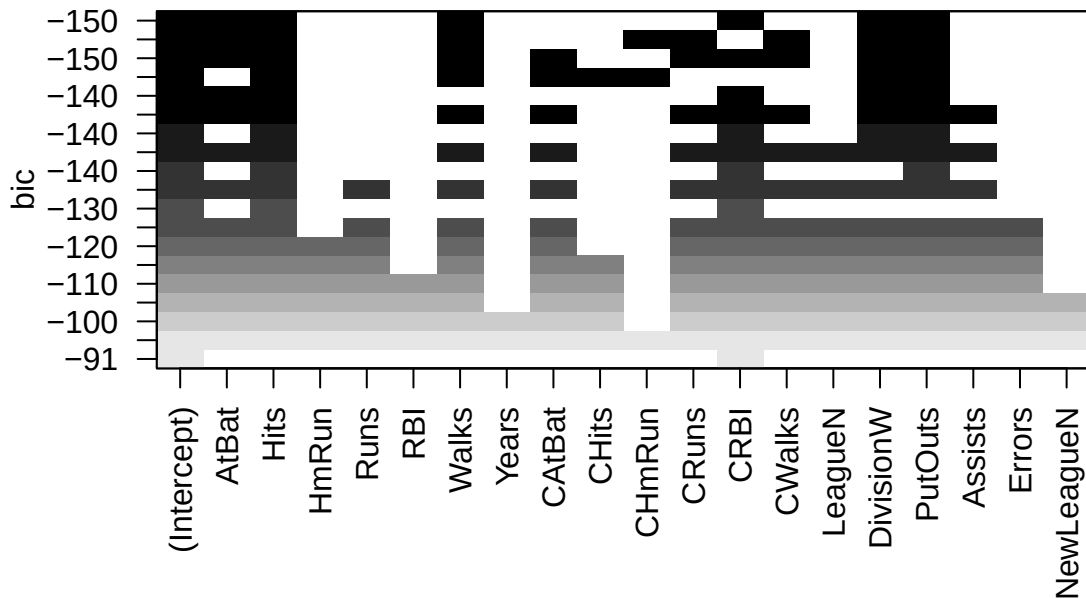
```
plot(regfit.full,scale="adjr2")
```



```
plot(regfit.full,scale="Cp")
```

```
plot(regfit.full,scale="bic")
```



```
coef(regfit.full,6)
```

```
## (Intercept)      AtBat      Hits      Walks      CRBI      DivisionW
##  91.5117981   -1.8685892   7.6043976   3.6976468   0.6430169  -122.9515338
##      PutOuts
##    0.2643076
```

Forward and Backward Stepwise Selection

```
regfit.fwd=regsubsets(Salary~.,data=Hitters,nvmax=19,method="forward")
summary(regfit.fwd)
```

```
## Subset selection object
## Call: regsubsets.formula(Salary ~ ., data = Hitters, nvmax = 19, method = "forward")
## 19 Variables (and intercept)
##           Forced in Forced out
## AtBat      FALSE      FALSE
## Hits       FALSE      FALSE
## HmRun      FALSE      FALSE
## Runs       FALSE      FALSE
## RBI        FALSE      FALSE
## Walks      FALSE      FALSE
```

```

## Years          FALSE      FALSE
## CatBat         FALSE      FALSE
## CHits          FALSE      FALSE
## CHmRun         FALSE      FALSE
## CRuns          FALSE      FALSE
## CRBI           FALSE      FALSE
## CWalks         FALSE      FALSE
## LeagueN        FALSE      FALSE
## DivisionW      FALSE      FALSE
## PutOuts        FALSE      FALSE
## Assists        FALSE      FALSE
## Errors         FALSE      FALSE
## NewLeagueN     FALSE      FALSE
## 1 subsets of each size up to 19
## Selection Algorithm: forward
##      AtBat Hits HmRun Runs RBI Walks Years CatBat CHits CHmRun CRuns CRBI
## 1 ( 1 ) " " " " " " " " " " " " " " " " " " " "
## 2 ( 1 ) " " "*" " " " " " " " " " " " " " " "*"
## 3 ( 1 ) " " "*" " " " " " " " " " " " " " " "*"
## 4 ( 1 ) " " "*" " " " " " " " " " " " " " " "*"
## 5 ( 1 ) "*" "*" " " " " " " " " " " " " " " "*"
## 6 ( 1 ) "*" "*" " " " " " " "*" " " " " " " " " "*"
## 7 ( 1 ) "*" "*" " " " " " " "*" " " " " " " " " "*"
## 8 ( 1 ) "*" "*" " " " " " " "*" " " " " " " "*" "*"
## 9 ( 1 ) "*" "*" " " " " " " "*" " " "*" " " " " "*" "*"
## 10 ( 1 ) "*" "*" " " " " " " "*" " " "*" " " " " "*" "*"
## 11 ( 1 ) "*" "*" " " " " " " "*" " " "*" " " " " "*" "*"
## 12 ( 1 ) "*" "*" " " "*" " " "*" " " "*" " " " " "*" "*"
## 13 ( 1 ) "*" "*" " " "*" " " "*" " " "*" " " " " "*" "*"
## 14 ( 1 ) "*" "*" "*" "*" " " "*" " " "*" " " " " "*" "*"
## 15 ( 1 ) "*" "*" "*" "*" " " "*" " " "*" "*" " " " "*" "*"
## 16 ( 1 ) "*" "*" "*" "*" "*" "*" " " "*" "*" " " " "*" "*"
## 17 ( 1 ) "*" "*" "*" "*" "*" "*" " " "*" "*" " " " "*" "*"
## 18 ( 1 ) "*" "*" "*" "*" "*" "*" "*" "*" "*" " " " "*" "*"
## 19 ( 1 ) "*" "*" "*" "*" "*" "*" "*" "*" "*" "*" " " "*" "*"
##      CWalks LeagueN DivisionW PutOuts Assists Errors NewLeagueN
## 1 ( 1 ) " " " " " " " " " " " "
## 2 ( 1 ) " " " " " " " " " " " "
## 3 ( 1 ) " " " " " " "*" " " " " "
## 4 ( 1 ) " " " " "*" "*" " " " " " "
## 5 ( 1 ) " " " " "*" "*" " " " " " "
## 6 ( 1 ) " " " " "*" "*" " " " " " "
## 7 ( 1 ) "*" " " "*" "*" " " " " " "
## 8 ( 1 ) "*" " " "*" "*" " " " " " "
## 9 ( 1 ) "*" " " "*" "*" " " " " " "
## 10 ( 1 ) "*" " " "*" "*" "*" " " " " "
## 11 ( 1 ) "*" "*" "*" "*" "*" " " " " "
## 12 ( 1 ) "*" "*" "*" "*" "*" " " " " "
## 13 ( 1 ) "*" "*" "*" "*" "*" "*" " " " "
## 14 ( 1 ) "*" "*" "*" "*" "*" "*" " " " "
## 15 ( 1 ) "*" "*" "*" "*" "*" "*" " " " "
## 16 ( 1 ) "*" "*" "*" "*" "*" "*" " " " "
## 17 ( 1 ) "*" "*" "*" "*" "*" "*" "*" "*"
## 18 ( 1 ) "*" "*" "*" "*" "*" "*" "*" "*"

```

```
## 19 ( 1 ) "*" "*" "*" "*" "*" "*" "
```

```
regfit.bwd=regsubsets(Salary~.,data=Hitters,nvmax=19,method="backward")
summary(regfit.bwd)
```

```
## Subset selection object
## Call: regsubsets.formula(Salary ~ ., data = Hitters, nvmax = 19, method = "backward")
## 19 Variables (and intercept)
##              Forced in Forced out
## AtBat          FALSE          FALSE
## Hits           FALSE          FALSE
## HmRun          FALSE          FALSE
## Runs           FALSE          FALSE
## RBI            FALSE          FALSE
## Walks          FALSE          FALSE
## Years          FALSE          FALSE
## CAtBat         FALSE          FALSE
## CHits          FALSE          FALSE
## CHmRun         FALSE          FALSE
## CRuns          FALSE          FALSE
## CRBI          FALSE          FALSE
## CWalks         FALSE          FALSE
## LeagueN       FALSE          FALSE
## DivisionW     FALSE          FALSE
## PutOuts        FALSE          FALSE
## Assists        FALSE          FALSE
## Errors         FALSE          FALSE
## NewLeagueN    FALSE          FALSE
## 1 subsets of each size up to 19
## Selection Algorithm: backward
##              AtBat Hits HmRun Runs RBI Walks Years CAtBat CHits CHmRun CRuns CRBI
## 1 ( 1 ) " " " " " " " " " " " " " " " " " " " "
## 2 ( 1 ) " " "*" " " " " " " " " " " " " " " "
## 3 ( 1 ) " " "*" " " " " " " " " " " " " " " "
## 4 ( 1 ) "*" "*" " " " " " " " " " " " " " " "
## 5 ( 1 ) "*" "*" " " " " " " "*" " " " " " " " "
## 6 ( 1 ) "*" "*" " " " " " " "*" " " " " " " " "
## 7 ( 1 ) "*" "*" " " " " " " "*" " " " " " " " "
## 8 ( 1 ) "*" "*" " " " " " " "*" " " " " " " "*"
## 9 ( 1 ) "*" "*" " " " " " " "*" " " " " " " "*"
## 10 ( 1 ) "*" "*" " " " " " " "*" " " " " " " "*"
## 11 ( 1 ) "*" "*" " " " " " " "*" " " " " " " "*"
## 12 ( 1 ) "*" "*" " " "*" " " "*" " " " " " " " "
## 13 ( 1 ) "*" "*" " " "*" " " "*" " " " " " " " "
## 14 ( 1 ) "*" "*" "*" "*" " " "*" " " " " " " " "
## 15 ( 1 ) "*" "*" "*" "*" " " "*" " " " " " " " "
## 16 ( 1 ) "*" "*" "*" "*" "*" "*" " " " " " " " "
## 17 ( 1 ) "*" "*" "*" "*" "*" "*" " " " " " " " "
## 18 ( 1 ) "*" "*" "*" "*" "*" "*" "*" " " " " " " "
## 19 ( 1 ) "*" "*" "*" "*" "*" "*" "*" " " " " " " "
##              CWalks LeagueN DivisionW PutOuts Assists Errors NewLeagueN
## 1 ( 1 ) " " " " " " " " " " " "
## 2 ( 1 ) " " " " " " " " " " " "
## 3 ( 1 ) " " " " " " "*" " " " " "
```

```
## 4 ( 1 ) " " " " " " " "
## 5 ( 1 ) " " " " " " " "
## 6 ( 1 ) " " " " " " " "
## 7 ( 1 ) "*" " " "*" " " " "
## 8 ( 1 ) "*" " " "*" " " " "
## 9 ( 1 ) "*" " " "*" " " " "
## 10 ( 1 ) "*" " " "*" " " " "
## 11 ( 1 ) "*" "*" "*" " " " "
## 12 ( 1 ) "*" "*" "*" " " " "
## 13 ( 1 ) "*" "*" "*" " " " "
## 14 ( 1 ) "*" "*" "*" " " " "
## 15 ( 1 ) "*" "*" "*" " " " "
## 16 ( 1 ) "*" "*" "*" " " " "
## 17 ( 1 ) "*" "*" "*" " " "*"
## 18 ( 1 ) "*" "*" "*" " " "*"
## 19 ( 1 ) "*" "*" "*" " " "*"
```

```
coef(regfit.full,7)
```

```
## (Intercept)      Hits      Walks      CatBat      CHits      CHmRun
## 79.4509472    1.2833513    3.2274264   -0.3752350    1.4957073    1.4420538
## DivisionW      PutOuts
## -129.9866432    0.2366813
```

```
coef(regfit.fwd,7)
```

```
## (Intercept)      AtBat      Hits      Walks      CRBI      CWalks
## 109.7873062   -1.9588851    7.4498772    4.9131401    0.8537622   -0.3053070
## DivisionW      PutOuts
## -127.1223928    0.2533404
```

```
coef(regfit.bwd,7)
```

```
## (Intercept)      AtBat      Hits      Walks      CRuns      CWalks
## 105.6487488   -1.9762838    6.7574914    6.0558691    1.1293095   -0.7163346
## DivisionW      PutOuts
## -116.1692169    0.3028847
```

Choosing Among Models

```
set.seed(1)
train=sample(c(TRUE,FALSE), nrow(Hitters),rep=TRUE)
test=(!train)
```

```
regfit.best=regsubsets(Salary~.,data=Hitters[train,],nvmax=19)
```

```
test.mat=model.matrix(Salary~.,data=Hitters[test,])
```

```

val.errors=rep(NA,19)
for(i in 1:19){
  coefi=coef(regfit.best,id=i)
  pred=test.mat[,names(coefi)]%*%coefi
  val.errors[i]=mean((Hitters$Salary[test]-pred)^2)
}

```

```
val.errors
```

```

## [1] 164377.3 144405.5 152175.7 145198.4 137902.1 139175.7 126849.0 136191.4
## [9] 132889.6 135434.9 136963.3 140694.9 140690.9 141951.2 141508.2 142164.4
## [17] 141767.4 142339.6 142238.2

```

```
which.min(val.errors)
```

```
## [1] 7
```

```
coef(regfit.best,10)
```

```

## (Intercept)      AtBat      Hits      HmRun      Walks      CAtBat
##  71.8074075  -1.5038124   5.9130470 -11.5241809   8.4349759  -0.1654850
##      CRuns      CRBI      CWalks  DivisionW      PutOuts
##   1.7064330   0.7903694  -0.9107515 -109.5616997   0.2426078

```

```

predict.regsubsets=function(object,newdata,id,...){
  form=as.formula(object$call[[2]])
  mat=model.matrix(form,newdata)
  coefi=coef(object,id=id)
  xvars=names(coefi)
  mat[,xvars]%*%coefi
}

```

```

# note change from test
regfit.best=regsubsets(Salary~.,data=Hitters,nvmax=19)
coef(regfit.best,7)

```

```

## (Intercept)      Hits      Walks      CAtBat      CHits      CHmRun
##  79.4509472   1.2833513   3.2274264  -0.3752350   1.4957073   1.4420538
##  DivisionW      PutOuts
## -129.9866432   0.2366813

```

```

k=10
set.seed(1)
folds=sample(1:k,nrow(Hitters),replace=TRUE)
cv.errors=matrix(NA,k,19, dimnames=list(NULL, paste(1:19)))

```

```

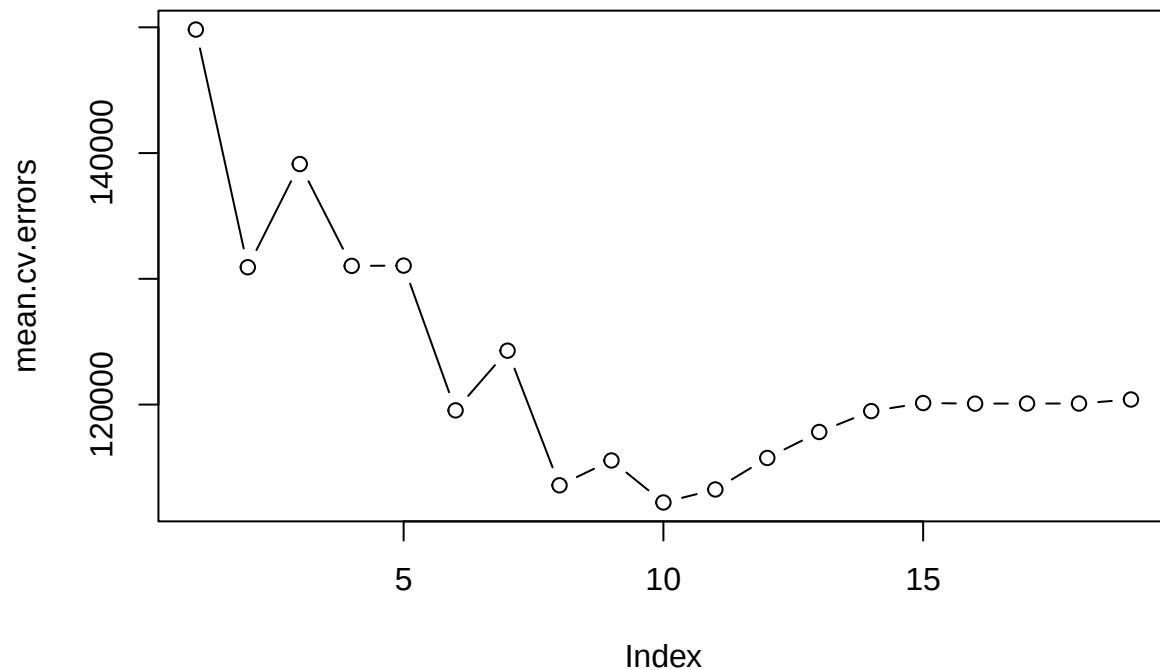
for(j in 1:k){
  best.fit=regsubsets(Salary~.,data=Hitters[folds!=j,],nvmax=19)
  for(i in 1:19){
    pred=predict(best.fit,Hitters[folds==j,],id=i)
    cv.errors[j,i]=mean( (Hitters$Salary[folds==j]-pred)^2)
  }
}

```

```
mean.cv.errors=apply(cv.errors,2,mean)
mean.cv.errors
```

```
##      1      2      3      4      5      6      7      8
## 149821.1 130922.0 139127.0 131028.8 131050.2 119538.6 124286.1 113580.0
##      9     10     11     12     13     14     15     16
## 115556.5 112216.7 113251.2 115755.9 117820.8 119481.2 120121.6 120074.3
##     17     18     19
## 120084.8 120085.8 120403.5
```

```
par(mfrow=c(1,1))
plot(mean.cv.errors,type='b')
```



```
# note change from text
reg.best=regsubsets(Salary~.,data=Hitters, nvmax=19)
coef(reg.best,10)
```

```
## (Intercept)      AtBat      Hits      Walks      CAtBat      CRuns
## 162.5354420 -2.1686501  6.9180175  5.7732246 -0.1300798  1.4082490
##      CRBI      CWalks  DivisionW      PutOuts      Assists
##  0.7743122 -0.8308264 -112.3800575  0.2973726  0.2831680
```